SE309.3 · SOFTWARE VERIFICATION & VALIDATION

සම්පූර්ණ විභාග අධ්‍යයන සටහන්

Complete Exam-Oriented Study Notes

Sinhala + English | Chapters 01-05

Lecture slide topics, definitions, examples, comparisons, process diagrams සහ විභාගයට ගැළපෙන පැහැදිලි කිරීම් එකට සකස් කර ඇත. [Additional Exam Point] ලෙස ලකුණු කළ කොටස් slide content එකට අමතරව අදාළ වන කරුණුය.

ප්‍රධාන මූලාශ්‍රය: user-provided SE309.3 lecture deck (269 slides). Slide content එකේ නොපැහැදිලි හෝ පරස්පර තොරතුරු සටහන් කර ඇත.

අන්තර්ගතය

1. V&V; Foundations: QA, terminology, coverage, risk, RTM

2. Advanced V&V;: planning, metrics, standards, reviews

3. Test-Driven Development (TDD)

4. Specialized testing: performance, API, security, white/black-box

5. Problem analysis, defect management, Agile/DevOps, AI/ML

Exam answer structure: Definition explanation English keywords scenario/example impact/benefit conclusion.

Chapter 01 - V&V; Foundations

Software Quality සහ SQA

Software Quality කියන්නේ software එක execute වීම පමණක් නොව, correct, reliable, secure, usable, maintainable සහ user needs වලට ගැළපීමයි. Banking app එකේ වැරදි මුදල් transfer වීම, hospital appointment නැතිවීම, GPA වැරදිව ගණනය වීම userට සැබෑ හානියක් කරයි.

Software Quality Assurance (SQA) යනු software process සහ product එක defined quality requirements සපුරන බව සහතික කිරීමට භාවිතා කරන planned, systematic ක්‍රියාමාර්ගයයි. Processes, reviews, standards, test planning/execution, defect tracking, metrics, process improvement සහ user satisfaction ඊට ඇතුළත්ය.

Verification සහ Validation

Aspect	Verification	Validation
ප්‍රශ්නය	"Are we building the product right?" Specification/design අනුව හරිද?	"Are we building the right product?" සැබෑ user need එකට ගැළපෙනවද?
අවධානය	Requirements, design, code, test docs වැනි work products.	Intended use, user workflow, business goals.
ක්‍රම	Reviews, inspections, walkthroughs, static analysis, code review.	Dynamic testing, UAT, beta/usability/field testing, stakeholder feedback.
උදාහරණය	Attendance requirement අනුව ID number දමා attendance mark කළ හැකිද?	Manual ID entry විශාල class එකකට lecturerට ප්‍රායෝගිකද; QR scan අවශ්‍යද?

Specification එක හරියට implement කළත් සැබෑ userට ප්‍රයෝජනවත් නොවේ නම් verification pass වුවත් validation fail විය හැක. දෙකෙම lifecycle පුරා අවශ්‍යය.

[Additional Exam Point] "Static = verification, dynamic = validation" මතක mnemonic එකක් පමණි; සම්පූර්ණ නියමයක් නොවේ. Unit test execute කළත් code එක specification එකට ගැළපෙනවාද බලන verification evidence විය හැක. අරමුණ අනුව classify කරන්න.

Static සහ Dynamic V&V;

Static	Dynamic
Software execute නොකර work product පරීක්ෂා කිරීම: requirement/design review, inspection, static analyzer.	Software run කර input ලබා දී actual result නිරීක්ෂණය කිරීම: unit සිට acceptance දක්වා tests.
Defects early සොයා අඩු වියදමින් වැළැක්විය හැක; runtime behavior නහවුරු නොකරයි.	Runtime behavior/failure පෙන්වයි; test data සහ environment අවශ්‍යය.

Testing philosophy: defect-free බව ඔප්පු කළ නොහැක

Testing defects සොයාගැනීමට සහ quality ගැන confidence වැඩි කිරීමටයි; software එකේ කිසිදු defect එකක් නැතැයි prove කිරීමට නොවේ. Exhaustive testing (සියලු inputs, paths, states, devices, timing combinations) සාමාන්‍යයෙන් කළ නොහැක. Test case තෝරාගැනීම, incomplete/wrong requirements, changing software සහ unexpected user behavior කාරණාගත tests pass වුවත් defects ඉතිරි විය හැක.

Login test එක valid credentials පමණක් නොව wrong/empty credentials, locked account, repeated attempts, long input, network interruption, password reset වැනි cases පරීක්ෂා කළ යුතුය. Good tester "කොහොමද pass වෙන්නේ?" පමණක් නොව "කොහොමද fail වෙන්නේ?" සොයයි.

Core QA terminology

Term	අර්ථය / උදාහරණය
Error	පුද්ගලයකුෙගේ වැරදි ක්‍රියාව හෝ අවබෝධය. Developer GPA formula එක වැරදිව තෝරුම්ගැනීම.
Defect / Fault / Bug	Requirement/design/code හි flaw. Total credits වනුවට subject count එකනේ බදෙන code.
Failure	Fault execute වීමනේ userට පනෙනෙ incorrect behavior. GPA 3.58 වනුවට 3.10 පනේවීම.
Test case	Objective/precondition, input/actions, expected result සහිත test specification.
Test data	Test සඳහා ලබාදනෙ values: student ID, marks, credits.
Expected / Actual	Expected = oracle/spec අනුව ලැබිය යුතුය; Actual = execution පසු ලැබුණේ.
Test oracle	Output නිවැරදිදැයි තීරණය කරන rule/source: approved formula, requirement, reference result.
Test suite / plan	Related test cases එකතුව / scope, levels, resources, schedule, criteria ඇතුළත් testing guide.
Traceability	Requirement design/code test defect links.

Cause chain: Human error software defect/fault execution under condition observed failure.

Test Coverage සහ Risk

Coverage මඟින් requirements, code, functions, risks හෝ scenarios වලින් testing කොපමණ ආවරණය කළාද පනේවයි. එය testing extent එකකි; software quality guarantee එකක් නොවේ.

• Requirement coverage = test-covered requirements ÷ total requirements × 100. 80/100 = 80%.

• Statement coverage: executable statements run වුණාද.

• Branch coverage: decision එකේ true/false (සියලු branches) run වුණාද.

• Condition coverage: compound decision එකේ එක් එක් Boolean condition true සහ false වුණාද.

• Path coverage: control-flow paths test කිරීම; loops නිසා path ගණන අධික විය හැක.

High code coverage වුණත් expected result assertion නොමැති නම් defect detection දුර්වලය. Requirement coverage 100% වුණත් performance/security requirement එක අර්ථ දක්වා නැත්නම් එය ආවරණය නොවේ.

Risk = Probability of failure × Impact of failure. Online exam හි timer, answer saving, auto-submit, login, result calculation high risk; theme color low risk. eChanneling හි payment, slot allocation, cancellation/refund, privacy, notification ඉදිරියනේ test කරන්න.

Requirement Traceability Matrix (RTM)

RTM එක requirements design/code modules, test cases, defects හා status සමඟ link කරයි. Missing coverage, change impact, retest scope සහ audit evidence හඳුනාගැනීමට උපකාරී වේ.

Requirement	Tests	Status	Action
REQ_01 Add/update/remove book and view total	TC_01-08	Passed	Covered
REQ_02 Fine calculation	TC_09-10	Partial	Add boundary/invalid cases
REQ_03 Refund	TC_11	Failed	Defect, fix, retest
REQ_04 SMS failure handling	-	Not tested	Coverage gap

Client fine rule වනෙස් කළහොත් RTM එකනේ අදාළ design, code, tests හඳුනාගත හැක. eChanneling case එකේ refund සහ SMS failure සඳහා test නොමැතිනම් “most tests passed” යන්න release readiness ඔප්පු නොකරයි.

Standards සහ lifecycle independence

Standards agreed terminology, process, documentation, evidence සහ quality expectations ලබාදයි. Slides සඳහන් කරන්නේ ISO/IEC/IEEE 29119 (testing concepts/process/documentation/design techniques) සහ IEEE 1012 (system/software/hardware V&V). Regulated work වල “tested” කියා පමණක් නොව, කුමක්/කවුරුන්/කෙසේ/කුමන requirement එකට/result කුමක්ද කියා පෙන්වීය යුතුය.

Waterfall, iterative, Agile, spiral, DevOps සියල්ලේම V&V; අවශ්යය. Waterfall හි phase reviews සහ final acceptance පැහැදිලි; Agile/DevOps හි repeated sprint reviews, automated checks සහ continuous user feedback වේ. Timing/style වෙනස් correctness, user fit, defect detection, risk reduction අරමුණ .

Chapter 01 Quick RevisionVerification = specification correctness; Validation = real-use suitability.Testing finds defects; it cannot prove none exist.Error defect/fault failure.Coverage shows extent, not quality; RTM links requirement and test evidence.Risk = likelihood × impact; prioritize high-risk functions.Keywords: SQA, V&V, static/dynamic, oracle, coverage, risk-based testing, RTM.
Possible questions: Compare V&V with attendance example; calculate requirement coverage; explain why 100% coverage is insufficient; classify GPA error/defect/failure.
[Additional Exam Point] State the coverage denominator explicitly (requirements, statements, branches, etc.).

Chapter 02 - Advanced V&V; Techniques

Objectives, constraints, trade-offs

Advanced V&V; යනු test execution පමණක් නොව early quality planning, defect prevention, suitable techniques, risk priority, metrics, traceability, compliance evidence සහ release confidence ය. Objectives: defect early detection, requirement/user-need satisfaction, risk reduction, reliability, release decision evidence, maintenance cost අඩු කිරීම.

Constraints: limited time/budget/resources, incomplete/changing requirements, complexity, limited domain knowledge/test data, release pressure. Exhaustive testing කළ නොහැකි නිසා deliberate prioritization අවශ්‍යය.

Trade-off	අර්ථය
Time vs Quality	Deep testing confidence වැඩි කරයි; delivery දිගු විය හැක.
Cost vs Coverage	More coverage සඳහා people/tools/effort වැය වේ.
Speed vs Documentation	Fast release evidence/records අඩු කළ හැක.
Automation vs Manual	Automation repeated tests වේගවත් කරයි; initial setup/maintenance අවශ්‍යයි; usability/exploration සඳහා human judgment වැදගත්.
Depth vs Deadline	Critical tests priority කර residual risk පැහැදිලිව පිළිගත යුතුය.

Safety-critical සහ regulated systems

Healthcare, aviation, railway, medical devices, banking, emergency response වැනි systems වල failure එක injury/death, money loss, privacy breach, legal issue හෝ trust damage කළ හැක. ඒ නිසා strict reviews, risk-based depth, independence, controlled change, traceability සහ documented approval අවශ්‍යය. Requirements reviewed, tests linked, defects tracked/fixed, results approved සහ release rationale recorded වීම audit evidence සපයයි.

V&V; Strategy, Test Plan, levels

V&V Strategy project-wide quality approach එකයි: verify/validate කළ යුතු දේ, roles/independence, test levels/types, tools, documents, metrics, risks, environment/data, automation, stop/release criteria. Test Plan execution guide එකයි: scope, in/out-of-scope features, levels/types, environment, data, responsibilities, schedule, risks, entry/exit criteria.

Level	Focus	eChanneling example
Unit	Function/class	Appointment fee calculation
Integration	Modules/interface interaction	Booking + payment gateway
System	Whole integrated product	Patient end-to-end booking
Acceptance	User/business suitability	Hospital staff validate actual workflow

Independence: developer self-test peer review separate QA external team client acceptance. Independent testers may challenge developer assumptions; early developer tests still provide fast feedback.

Documents, reviews, static analysis

Artifacts: V&V; plan, test strategy/plan, RTM, review reports, test cases/data, execution report, defect report, test summary, release recommendation. Defect report includes ID, title, exact description/steps, expected/actual, severity/priority, evidence, assignee/status.

Reviews inspect requirements, designs, source, test cases, manuals, schemas and interfaces without execution. Walkthrough = author explains; technical review = experts check correctness; inspection = formal roles/checklist; peer review = team members; management review = progress/risk/readiness.

Requirement checklist: clear, complete, consistent, testable, feasible, necessary, traceable; business rules and non-functional requirements present? “System should be fast” is vague. “Doctor search results shall display within 3 seconds” is measurable/testable.

Static analysis checks code without running: syntax, unused variables, duplicate code, complexity/code smells, possible null errors, security/dependency problems, coding standards. Benefits: early discovery, less later cost, maintainability, CI gate. Limits: false positives, misses runtime defects, cannot fully validate user needs/usability, cannot replace dynamic

testing. Slide examples: SonarQube, ESLint, PMD, Checkstyle, SpotBugs.

Metrics and their limits

• Test cases prepared/executed; pass/fail percentage; defect count.

• Requirement/code coverage; defect density; defect leakage; defect removal efficiency; mean time to fix.

• Defect density = defects ÷ software size. 20 defects / 10,000 LOC = 2 defects per 1,000 LOC.

• Defect leakage = defects escaping a test phase or reaching production; can signal weak test design, missing coverage, unrealistic data or regression gaps.

Metrics must be interpreted in context: 100% execution/high coverage/large test count/low defect count alone does not prove quality. Low defect count may indicate weak testing, not excellent software.

Lifecycle-wide V&V; and QA mindset

Phase	V&V; actions
Requirements	Review clarity, completeness, consistency, testability; validate acceptance criteria.
Design	Review architecture, DB, security, UI flow.
Implementation	Code review, unit test, static analysis.
Testing	Integration/system/regression/performance/security/acceptance.
Maintenance	Change impact, retest, regression, user feedback/monitoring.

Agile: story/acceptance criteria review, sprint testing, feedback and regression. DevOps: automated tests, CI checks, deployment gates and production monitoring. QA skills: requirement analysis, test design, manual/automation/API/DB, defect reports, programming basics, version control, CI/CD and communication. Mindset: critical, curious, user/risk-focused, ethical, precise, questions assumptions, learns continuously.

Chapter 02 Quick RevisionStrategy = project quality approach; Plan = test execution guide.V&V applies requirements through maintenance; not only final test stage.Regulated quality needs evidence/traceability as well as tests.Static checks early code; it cannot substitute runtime/user testing.Metrics support decisions; never treat one number as a quality verdict.Keywords: safety-critical, compliance, independence, test levels, artifacts, defect leakage, review, static analysis.
Exam questions: Write strategy for eChanneling; distinguish strategy/plan; static-analysis benefits/limits; critique misleading metrics.

Chapter 03 - Test-Driven Development (TDD)

Definition සහ Red-Green-Refactor

Test-Driven Development (TDD) යනු production code ලිවීමට පෙර test case එක ලියා, එය pass කිරීමට අවශ්‍ය code පියවරනේ පියවර නිර්මාණය කරන development approach එකකි. Test එක expected behavior පැහැදිලි කරන නිසා TDD testing technique එකක් පමණක් නොව design discipline එකකි.

1. Red: කුඩා behavior test එක ලියා run කරන්න; implementation නැති/අසම්පූර්ණ නිසා fail විය යුතුය. Fail හේතුව තහවුරු කරන්න.

2. Green: Test pass කරන minimum production code පමණක් ලියන්න.

3. Refactor: Tests pass තබාගනෙ structure/duplicate/complexity පිරිසිදු කරන්න; නැවත tests run කරන්න.

4. ඊළඟ behavior එක සඳහා cycle repeat කරන්න.

> Flow: Requirement/behavior → failing test (Red) → minimum code (Green) → safe refactor → regression run → next behavior.

Grade calculator example සහ slide ambiguity

Marks-to-grade function එකකට valid range, pass boundary, grade-band boundaries, negative සහ maximumට ඉහළ invalid marks සඳහා tests ලියන්න. Oracle එක approved grading rule එක විය යුතුය. Slide එකක 0-39 fail, 40-75 pass ලෙස දී 76-100 පැහැදිලි නොකරයි; වෙනත් slide එකක 75-100=A, 65-74=B, 50-64=C, below 50=F ලෙස වෙනත් bands ඇත. මයෙ lecture slides තුළ අසංගත/අසම්පූර්ණයි. Exam scenario එකේදී ඇති rule එක පමණක් යොදාගන්න; missing interval අනුමාන නොකරන්න. Code example එක invalid range එකට F දීමත් specification එකනේ තහවුරු කළ යුතුය.

> [Additional Exam Point] Pass mark 50 නම් 49, 50, 51 වැනි boundary values test කරන්න. Expected grade/valid range එක business rule එකනේ ගන්න.

Benefits, limits, good unit tests

Benefits	Challenges
Behavior early පැහැදිලි; regression suite ඉක්මනින්; early defects; modular/testable design; refactor confidence.	Initial effort/skill; legacy/GUI/distributed behavior අපහසු; changing requirements නිසා test maintenance; weak tests false confidence දයි.

Good unit tests: fast, independent, repeatable, clear, focused, automated, meaningful. Test එකක් එක් ප්‍රධාන behavior එකක් check කර expected result assert කළ යුතුය; code run කළ බව පමණක් නොපනේවිය යුතුය.

TDD mindset: expected behavior මොකක්ද? Failure paths? Boundaries? Invalid input? Future change එකනේ behavior බිඳේද? Regression tests නව code නිසා පැරණි behavior කැඩීම ඉක්මනින් පනේවයි.

> Chapter 03 Quick RevisionRed (fail) → Green (minimum pass) → Refactor (same behavior, cleaner design).TDD test-first development/design discipline එකකි.Tests behavior අර්ථ දක්වයි; expected output සඳහා oracle එක අවශ්‍යය.Lecture grade ranges conflicting; exam prompt rule එක භාවිතා කරන්න.Keywords: TDD, Red-Green-Refactor, unit test, boundary, oracle, regression.
> Exam questions: Explain TDD with grade function; discuss benefits/limits; design boundary tests for stated pass rule.

Chapter 04 - Specialized Testing Approaches

Performance testing

Performance testing මඟින් workload යටතේ response time, throughput, stability සහ capacity මනින non-functional testing එකකි. Functional test අසන්නේ booking සිදු වුණාද; performance test අසන්නේ users 3,000ක් එකවරදී service එක කෙසේද යන්නයි.

Type	අරමුණ	Example
Load	Expected/peak workload යටතේ behavior මැනීම.	Mock exam එකට 3,000 students.
Stress	Normal capacity ඉක්මවා breaking point, graceful degradation/recovery බලන්න.	Concurrent users වැඩිකර service failure/recovery සොයන්න.
Scalability	Load/data වැඩිවන විට capacity ප්‍රමාණවත් ලෙස scale වේද.	Users 1,000 සිට 10,000 දක්වා වැඩිවීම.
Endurance/Soak	දිගු sustained load යටතේ stability/resource leak.	පැය ගණනක online examination.

“Fast” වනුවට user load, response-time threshold, throughput, duration වැනි measurable non-functional criteria අවශ්‍යය. Functional tests පමණක් concurrency/network delays දිගුකාලීන instability නොපනේවයි.

API testing

API testing UI හරහා නොව service request/response boundary එකේ validation, business rules, status/body/schema, auth, error handling, persistence, duplicate requests, timeout සහ integration පරීක්ෂා කරයි. UI tests pass වුවත් API integration fail විය හැක.

Status	අර්ථය	Example
200 OK	සාර්ථක response	Student login data ලැබුණි
201 Created	Resource නිර්මාණය	Appointment booking
400 Bad Request	Input invalid/missing	Patient ID/date වැරදියි
401 Unauthorized	Authentication නැත/අසාර්ථක	Login නොවූ request
403 Forbidden	Identity ඇත; permission නැත	Patient admin report ඉල්ලයි
404 Not Found	Resource හමු නොවීය	Invalid doctor ID
500 Server Error	Unexpected server failure	Database/gateway failure

Important cases: invalid body/missing field, payment gateway down, timeout, duplicate request, unauthorized access. Payment fail නම් booking confirm නොකරන්න; refund success නම් notification path ද පරීක්ෂා කරන්න. Error message පැහැදිලි විය යුතු නමුත් secret/internal stack නොපනේවිය යුතුය.

Security testing

Unauthorized access/disclosure/modification/destruction වලින් data/functions ආරක්ෂිතද යන්න ඇගයීම. Marks, money, medical records, passwords වැනි data ඇති systems සඳහා අත්‍යවශ්‍යය.

• Authentication: “Who are you?” identity/login/credential/reset behavior.

• Authorization: “What may you access/do?” role permissions, least privilege.

• Input validation: Untrusted/invalid values check/sanitize කර injection/abuse paths පරීක්ෂා කිරීම.

• Session management: expiry, logout, token/session handling.

• Data protection/access control: at-rest/in-transit encryption; passwords plaintext නොවීම; sensitive info URL/log වල නොපනේවීම; patient/student data isolation.

Patientට වනෙත් patientගේ record නොපනෙය යුතුය; studentට marks වනෙස් කිරීමට නොහැකි විය යුතුය. Functionally correct app එකක් authorization bypass නිසා insecure විය හැක.

White-box vs Black-box

Aspect	Black-box	White-box
Focus	Inputs/outputs, observable user behavior.	Internal code, logic, branches, data flow.
Code knowledge	අවශ්‍ය නැත.	අවශ්‍යය.
Users	QA, business users, clients.	Developers, technical testers.
Techniques	Equivalence partitioning, BVA, decision table, state/use-case.	Statement/branch/condition/path/loop/data flow, static analysis.
Login example	Valid/invalid credentials වල visible result.	Validation logic branches සහ internal path.

White-box coverage techniques

• Statement coverage: executable statements සියල්ල run වුණාද; logic නිවැරදි බව නොකියයි.

• Branch coverage: decision outcome සියල්ල (true/false, alternatives) execute වුණාද.

• Condition coverage: compound decision එකේ එක් එක් Boolean condition true සහ false වුණාද.

• Path coverage: control-flow paths; loop නිසා exhaustive path testing ප්‍රායෝගික නොවීය හැක.

• Loop testing: 0, 1, typical, max iterations සහ loop boundary/termination cases.

• Data-flow testing: variables define/use/update paths; uninitialized හෝ වැරදි data propagation සවීම.

[Additional Exam Point] Branch coverage තිබුණත් සෑම condition combination එකක්ම හෝ execution path එකක්ම cover වීම සහතික නැත. Coverage criteria වනෙස් denominator සහ strength ඇති බව පිළිතුරේ සඳහන් කරන්න.

Black-box test design

• Equivalence Partitioning: සමාන හැසිරීමක් ඇති input classes වලින් representative value තෝරන්න. Age 18-60 valid නම් below, valid range, above range classes.

• Boundary Value Analysis: min/max සීමා අසල defects වැඩි; min−1, min, min+1, max−1, max, max+1 test කරන්න.

• Decision table: conditions × outcomes matrix. Payment success? Slot available? - confirm/reject/refund combinations.

• State transition: states, events, valid/invalid transitions. Exam ScheduledActiveSubmittedMarked; time expiry auto-submit.

• Use case testing: actor goal end-to-end workflow and alternate/error flows: patient searchbookpayconfirm.

Integration, system, usability, Agile/DevOps

Integration module/API interactions පරීක්ෂා කරයි; payment fail response එක booking status update නොකරන බව බලයි. System complete integrated product එක functional සහ non-functional requirements සමඟ පරීක්ෂා කරයි. Usability representative usersට tasks පහසුවනේ/නිවැරදිව කළ හැකිද බලයි; technically correct fund transfer UI එක users confuse කර වැරදි approve කරවිය හැක.

Agile තුළ සෑම sprint එකකම story/acceptance review, unit/TDD, code review, exploratory/acceptance, regression tests වේ. DevOps pipeline සාමාන්‍යයනේ build verificationunitintegration/APIstatic/security scancoverage/regressiondeployment check/smokemonitoring. Critical tests fail වුවහොත් pipeline නවතී. Smoke යනු basic critical build health; Regression යනු changes නිසා existing behavior බිඳීද.

Chapter 04 Quick RevisionLoad=expected load; Stress=beyond limit; Scalability=growth handling; Endurance=long duration.Authentication=identity; Authorization=permissions.Black-box=external behavior; White-box=internal structure.EP classes; BVA boundaries; decision table rules; state transitions; use cases/workflows.Functional pass පමණක් release readiness නොවේ: performance, security, usability, resilience ද අවශ්‍යය.Keywords: API, load/stress/scalability/endurance, authentication, branch/condition, EP, BVA, CI/CD, smoke, regression.
Exam questions: Compare performance types; design API failure tests; EP/BVA for input rule; compare black/white-box; outline online-exam CI pipeline.

Chapter 05 - Problem Analysis & Defect Management

Problem analysis: complaint actionable defect

Problem analysis යනු මොකක් වැරදුණාද, කොතැනද, ඇයි, නැවත නොවීමට කුමක් කළ යුතුද සොයාගැනීමයි. Problem එක code bug එකක් පමණක් නොව requirement/design/data/configuration/integration/process issue එකක් විය හැක. “Exam system is not working” vague ය; “network reconnect පසු active exam timer 60 minutes ලෙස reset වේ” actionable ය.

1. Complaint/incident එක පැහැදිලිව ලියන්න.

2. එකම role/data/device/browser/build/network සමඟ reproduce කරන්න.

3. Screenshots, logs, timestamps, API response, test data එකතු කරන්න.

4. Module/integration point isolate කරන්න.

5. Expected/actual, steps, impact, severity/priority සමඟ defect report සාදන්න.

Defect report සහ lifecycle

Report fields: ID, concise title, environment/build, description, exact steps, expected vs actual, severity/priority, evidence, assignee/status. Timer example: start 60-minute exam; 20 minutes පසු network 2 minutes disconnect; reconnect; expected ~38 minutes, actual 60/incorrect. Exam fairness බලපාන නිසා severity/priority high.

State	Action
New	Tester/user/monitor report කරයි.
Triaged	Validity, impact, urgency, owner, release effect assess.
Assigned	Developer/team investigates.
Fixed	Code/config/data correction complete.
Retest	Original reproduction steps නැවත verify.
Regression	Related workflows අලුත් fix එකනේ කැඩී නැද්ද check.
Closed / Reopened	Pass නම් close; issue තවම පවතිනවා නම් evidence සමඟ reopen.

Severity vs Priority

Severity	Priority
Technical/user impact කොපමණද?	Fix කිරීම කොතරම් urgent ද?
Impact-based classification.	Release/business scheduling decision.
Payment charged but booking absent: high.	Official graduation certificate typo: low severity වුවත් releaseට පෙර high priority.

Rare data corruption in unused legacy module = high severity/low priority විය හැක. Minor icon = low/low. දෙකෙ එකම දෙයක් නොවේ.

Debugging, fault isolation, root cause

Debugging defect root cause හඳුනා fix කිරීම; fault isolation failure එක exact module/function/API/DB/config/environment වනෙ සීමා කිරීම. Evidence මත hypothesis එක බැගින් පරීක්ෂා කරන්න.

• Binary search: components/changes බදො failure ආරම්භය සොයන්න.

• Log analysis: timestamps, error IDs, request sequence.

• API tracing: services අතර request-response follow.

• Database verification: insert/update/rollback නිවැරදිද.

• Environment/version comparison: local/staging, recent commit/config වනෙස්කම්.

• Stack traces, breakpoints, controlled reproduction ද භාවිතා කරන්න.

Root Cause Analysis (RCA) symptom එකට පමණක් patch නොදා underlying cause සොයයි. Methods: Five Whys, fishbone, causal chain, defect clustering, Pareto analysis.

Five Whys - payment failed but booking confirmed
1. Booking status payment responseට පරෙ update විය. 2. Booking/payment APIs transaction ලෙස coordinate නොවීය. 3. Error path rollback නොකරයි. 4. Requirement failure behavior පැහැදිලි නැත. 5. Decision table/integration test payment failure ආවරණය නොකළේය.
Root causes: requirement ambiguity + missing rollback/error-path testing. Code fix පමණක් නොව requirement/test process ද නිවැරදි කරන්න.

Defect clustering: defects කිහිපයක් modules කීපයක එකතුවිය හැක; historical defect data අනුව billing/pharmacy වැනි hotspots වල regression/review වැඩි කරන්න. Pareto heuristic එක rigid universal 80/20 law එකක් ලෙස නොගන්න.

Defect metrics සහ tracking tools

Defect count, density, leakage, removal efficiency, mean time to detect/fix, reopen rate. Trends තේරුම්ගනෙ quality decision ගන්න; raw count එකනේ තනිව team/product quality තීරණය නොකරන්න. Tools: Jira, Azure DevOps, GitHub/GitLab Issues, Bugzilla, YouTrack, TestRail integration. Workflow, severity/priority, attachments, requirement/test links, dashboards, notifications, audit history සහ CI results support කරයි.

Agile, DevOps, continuous testing සහ AI/ML

Agile QA සෑම sprint එකකම: story/acceptance review, test design, exploratory/sprint/regression. Defects backlog එකේ risk අනුව prioritize කරයි. Definition of Done තුළ unit tests, code review, acceptance criteria, QA verification, regression checks තිබිය හැක; recurring defects retrospective එකක process change කරයි.

DevOps “shift left” unit/static/security checks early කරයි; “shift right” monitoring/logs/alerts/user feedback මඟින් production quality බලයි. CI/CD commit/deploy එකකට build, unit, integration, API, static/security, coverage, regression, smoke tests run කර critical failure එකකදී deployment නවත්වයි. Stable, frequent, repeatable, high-risk tests පළමුව automate කරන්න; unclear/changing features early automate කළොත් maintenance cost වැඩි විය හැක.

Slides AI/ML in testing, defect prediction, prevention හඳුන්වයි. Historical data මඟින් risky modules හඳුනාගැනීම, tests prioritize/generate කිරීමට උදව් විය හැක. [Additional Exam Point] Generated tests/results human review සහ correct oracle මඟින් validate කරන්න; biased/incomplete data, false positives, privacy/explainability අවදානම්. AI professional judgment/requirements ownership/security review වනුවට නොවේ.

Scenario application

• Online exam: timer reset/answer loss-browser/network logs, timestamps, request IDs, reconnect/concurrency tests; fairness නිසා high risk; regression timer/save/auto-submit/results.

• Hospital: double booking, wrong bill, record access, lab mismatch, stock update-role, transaction, API, DB, audit log check; privacy/clinical accuracy top priority.

• Food delivery discount: discount+delivery total error-acceptance rule for stacking/tax/rounding; unit/API/invoice/payment regression; retrospective RCA.

Chapter 05 Quick RevisionActionable report = environment + exact steps + expected/actual + impact + evidence.Newtriageassignedfixedretestregressionclosed/reopened.Severity=impact; Priority=urgency.RCA underlying cause සොයයි; fix පසු retest සහ regression.Continuous testing pipeline critical failure එකක release block කරයි; post-release feedback ද වැදගත්.Keywords: incident, fault isolation, RCA, Five Whys, severity, priority, leakage, regression, CI/CD, defect prediction.
Exam questions: Write timer defect report; severity vs priority; Five Whys payment case; propose CI/CD tests for refund; AI/ML benefits and risks.

අවසාන විභාග පුනරාවර්තනය

Must-remember definitions

• Verification: work products defined conditions/specification සපුරනවාද ඇගයීම.

• Validation: product intended use සහ user needs සපුරනවාද ඇගයීම.

• Test coverage: tests මඟින් requirements/code/risk/scenarios කොපමණ ආවරණය කළාද.

• RTM: requirement එක design/code/test/defect artifacts සමඟ link කරන matrix.

• TDD: test-first development; Red-Green-Refactor.

• Severity/Priority: defect impact / fix urgency.

• RCA: defect නැවත ඇතිවීම වැළැක්වීමට underlying cause හඳුනාගැනීම.

High-value comparisons

Comparison	Memory cue
Verification vs Validation	Specification correctness vs real-use suitability.
Static vs Dynamic	Execution නැති review/analyzer vs software run කර test කිරීම.
Black-box vs White-box	External behavior vs internal structure.
Authentication vs Authorization	Who are you? vs what may you do?
Load / Stress / Scalability / Endurance	Expected load / beyond limit / growth / sustained duration.
Severity vs Priority	Impact / urgency.
Smoke vs Regression	Basic build health / change එකනේ old behavior බිඳීද.
Defect detection vs prevention	Test finding / early review, clear requirement, static analysis.

Integrated exam questions (brief answer plans)

1. Online exam timer network reconnect පසු reset වේ. Lifecycle-wide V&V යෝජනා කරන්න.
Requirements වල time authority/offline behavior measurable කරන්න; design review; timer unit tests; API/integration reconnect tests; network/load scenarios; UAT fairness; RTM; evidence-rich defect report; severity/priority; fix retest/regression; release gate.

2. eChanneling UI booking success පනේවයි නමුත් payment fail. Analyze.
API/integration/error-path gap; payment/slot decision table; rollback/idempotency; Five Whys; high impact; RTM update; fix retest plus booking/refund/notification regression; monitor.

3. Why doesn't 100% coverage prove quality?
Weak/no assertions, poor oracle, incomplete/wrong requirement, missing NFR/security/usability/performance, network/environment scenarios; coverage is extent only; combine risk-based design, traceability, exploratory/UAT.

4. Hospital management system test strategy?
Rank clinical correctness/privacy/billing/booking; unit/integration/system/acceptance; white-box branches/data flow; black-box EP/BVA/state/use-case; role access, load, static review; RTM, defect evidence and regression.

5. GPA/marks calculator test cases?
First establish formula/range as oracle; valid/invalid equivalence classes; boundaries; weighted credits; zero credits/division handling; grade-band edges; compare expected/actual; TDD; explain errordefectfailure.